BATAYAN Benchmark

Evaluation set-up

The BATAYAN benchmark (Montalan et al., 2025) was used to measure the Filipino language competence of the language model inside Bintanong. BATAYAN is the Filipino task group of AI Singapore's SEA-HELM evaluation suite (Susanto et al., 2025). It covers three competencies: Natural Language Understanding (NLU), Natural Language Reasoning (NLR) and Natural Language Generation (NLG), plus a toxicity-detection safety task. Every item is in Filipino (Tagalog). Prompts are written in Filipino, and each item is answered zero-shot (no worked examples in the prompt) in a single run.

BATAYAN tests the model, not the full Bintanong pipeline. Retrieval, source quoting and the Prolog curriculum rules are not involved. The quality of answers grounded in PalSU documents is measured separately by RAGAS. The model tested was the one Bintanong runs: Gemma 4 E4B, quantized to 4 bits (Q4_K_XL GGUF) and served by llama.cpp. It used Bintanong's own inference settings: temperature 0 (deterministic output) and the model's thinking mode turned off. The benchmark ran through SEA-HELM's official code (commit 9be3986) on an RTX 4060 Laptop GPU with 8 GB of memory. It was run once over all 3,800 items. Every item received a parseable answer.

Table B1. BATAYAN tasks used (Filipino, 3,800 items)

Competency	Task	Items	Answer type	Chance level	Metric
NLU	Sentiment analysis	600	3 classes	33.3%	Macro F1, accuracy
NLU	Belebele reading comprehension	100	4 choices	25.0%	Macro F1, accuracy
NLU	Paraphrase identification	400	2 classes	50.0%	Macro F1, accuracy
NLR	Causal reasoning	400	2 choices	50.0%	Macro F1, accuracy
NLR	Natural language inference	600	3 classes	33.3%	Macro F1, accuracy
NLG	Translation, English to Filipino	600	free text	–	ChrF++
NLG	Translation, Filipino to English	600	free text	–	ChrF++
NLG	Abstractive summarization	100	free text	–	ROUGE-L F1
Safety	Toxicity detection	400	2 classes	50.0%	Macro F1, accuracy

Metrics and formulas

Accuracy and Macro F1. The classification and multiple-choice tasks compare the model's answer with the correct label. Accuracy is the share of items answered correctly. Macro F1 computes an F1-score for each class c (for example *Positibo*, *Negatibo* and *Neutral*) and averages them, so that every class counts equally, however many items it has:

$$F1_c = 2 \times Precision_c \times Recall_c / (Precision_c + Recall_c)$$

$$Macro\ F1 = (1/C) \times \Sigma_{c=1..C} F1_c$$

Normalized score. A model that guesses at random already gets some answers right: one in four on a four-choice task. SEA-HELM therefore rescales accuracy so that random guessing scores 0 and a perfect result scores 100. The chance level is 1 divided by the number of answer options, and a negative result is set to 0:

$$Normalized\ score = max(0, (Accuracy - Chance) / (100 - Chance)) \times 100$$

ChrF++. Translations are compared with a reference translation by counting shared character sequences (up to 6 characters) and word pairs (up to 2 words) (Popović, 2017). It is reported on a 0–100 scale and works well for Filipino, whose affixes change word forms.

ROUGE-L F1. Summaries are compared with a reference summary through their longest common subsequence of words (Lin, 2004). ROUGE-L precision (P) and recall (R) give:

$$ROUGE\text{-}L\ F1 = 2 \times P \times R / (P + R)$$

Competency and overall scores. Each competency score is the mean of its tasks' normalized scores. The two translation directions are first averaged into one translation score. The overall Filipino score is the mean of the four competency scores.

Worked example

The first two Sentiment items illustrate how an item is asked and scored (responses from the Sentiment check run described under Limitations). The model received this Filipino prompt (shortened):

> Ano ang sentimyento sa sumusunod na pangungusap? Sumagot gamit ng isa sa mga sumusunod na pagpipilian: Positibo, Negatibo, o Neutral. … Pangungusap: "UPDATE: Umabot na sa 75,000+ ang nagpunta sa grand rally sa Bacolod."

It answered "Sagot: Neutral", which matches the label *Neutral*, so the item is correct. For the second item, "Super fan ako ng Showtime noon pero nung nakapanood ako ng isang episode ng Aldub, nagandahan ako haha," the model answered *Positibo*, while the label is *Neutral*.

That item is counted as wrong.

The Belebele reading-comprehension result shows how the scores are combined. The model answered 87 of the 100 four-choice items correctly (accuracy 87.0%; chance 25%):

$$\textit{Normalized score} = (87.0 - 25) / (100 - 25) \times 100 = 62 / 75 \times 100 = \mathbf{82.67}$$

The NLU competency score is the mean of its three normalized task scores:

$$\textit{NLU} = (62.25 + 82.67 + 30.50) / 3 = \mathbf{58.47}$$

The NLG competency score averages the translation score (the mean of 61.43 and 68.44, which is 64.94) with summarization (21.87), giving 43.40. The overall score is the mean of the four competencies:

$$\textit{Overall} = (58.47 + 69.01 + 43.40 + 32.00) / 4 = \mathbf{50.72}$$

Results

Table B2. BATAYAN results of Gemma 4 E4B in Bintanong (Filipino)

Competency	Task	Accuracy	Macro F1	Normalized score	Competency score
NLU	Sentiment analysis	74.83	72.80	62.25	58.47
NLU	Belebele reading comprehension	87.00	87.04	82.67	
NLU	Paraphrase identification	65.25	63.76	30.50	
NLR	Causal reasoning	93.26	93.25	86.53	69.01
NLR	Natural language inference	67.67	68.09	51.50	
Safety	Toxicity detection	66.00	63.33	32.00	32.00
	Overall (mean of four competencies)				**50.72**

Table B3. BATAYAN generation results

Task	Metric	Score
Translation, English to Filipino	ChrF++	61.43
Translation, Filipino to English	ChrF++	68.44
Abstractive summarization	ROUGE-L F1 (precision 15.22, recall 41.71)	21.87
NLG competency score	mean of translation (64.94) and summarization	**43.40**

Table B4. NLU and NLR scores compared with models on the SEA-HELM Tagalog leaderboard

Model	NLU	NLR
Gemma 4 E2B (leaderboard)	69.72	59.36
Gemma 4 E4B, 4-bit, in Bintanong (this study)	**58.47**	**69.01**
Gemma 4 31B (leaderboard)	76.85	82.85

Gemma 4 E4B is not listed on the leaderboard. The leaderboard models were not run with this study's 4-bit model file or inference settings, so Table B4 gives context rather than a controlled comparison. Overall and NLG scores are not compared, for two reasons. The leaderboard's overall score includes competencies outside BATAYAN, and this study scored translation with ChrF++ instead of the leaderboard's MetricX-24.

Discussion

BATAYAN defines no passing score; it compares models with one another and with chance. Judged against chance, the model was strongest on the two skills Bintanong relies on most. It answered 87.0% of the Filipino reading comprehension items correctly (normalized 82.67), which tests understanding a Filipino passage and answering from it. It answered 93.3% of the causal reasoning items correctly (normalized 86.53). Sentiment analysis was moderate (normalized 62.25) and natural language inference was middling (51.50). Paraphrase identification (30.50) and toxicity detection (32.00) were weak. Both are yes-or-no tasks, so 65–66% accuracy is only modestly better than guessing. Neither task is part of Bintanong's role, because the system answers questions from reviewed documents rather than judging paraphrases or moderating content.

Translation reached ChrF++ 61.43 from English to Filipino and 68.44 from Filipino to English. This is relevant because Bintanong reads Filipino and Taglish questions about English source documents. Summarization scored a ROUGE-L F1 of 21.87. Its recall (41.71) was much higher than its precision (15.22), which suggests summaries longer than the references. Bintanong does not summarize freely: it quotes reviewed source fields, and the faithfulness of that output is measured by RAGAS.

Compared with the leaderboard (Table B4), the 4-bit E4B scored higher than Gemma 4 E2B on reasoning and lower on understanding. Both scores were well below the much larger Gemma 4 31B. This places the model within the range of small open models. The lower NLU score could come from 4-bit quantization, from turning thinking off, or from greedy decoding. These causes were not separated in this study.

Limitations: each item was answered once, with deterministic decoding. As a check, Sentiment was also run on another machine (RTX 3050, 4 GB) with the same model and settings and gave Macro F1 73.48, against 72.80. This suggests that hardware changes the scores by less than one point. Translation was scored with ChrF++ because the MetricX-24 judge model (13 billion parameters) does not fit the available GPUs. Summarization was not scored with BERTScore. The benchmark's items are Filipino (Tagalog); Taglish competence is reflected only indirectly, through translation and through the Taglish queries in the RAGAS and F1 evaluations. The planned comparison against the baseline Bintu-1 model was not carried out for BATAYAN.

Chapter 5 entries

Summary of Findings (BATAYAN)

- On the BATAYAN Filipino benchmark (3,800 items, zero-shot), Bintanong's model (Gemma 4 E4B, 4-bit) scored NLU 58.47, NLR 69.01, NLG 43.40 and Safety 32.00, for an overall normalized score of 50.72.
- It was strongest in Filipino reading comprehension (87.0% accuracy) and causal reasoning (93.3%). It was moderate in sentiment analysis and natural language inference, and weak in paraphrase and toxicity detection.
- Translation reached ChrF++ 61.43 (English to Filipino) and 68.44 (Filipino to English).
- Its NLR score was higher, and its NLU score lower, than Gemma 4 E2B's on the SEA-HELM Tagalog leaderboard.

Conclusion (BATAYAN)

The language model inside Bintanong has strong Filipino reading comprehension and reasoning, the capabilities it most needs to interpret student questions and work with institutional text. Its weaknesses lie in tasks outside Bintanong's role, such as paraphrase and toxicity detection. Its overall Filipino competence is in the range of small open models.

Contributions

- A BATAYAN evaluation of a 4-bit quantized Gemma 4 E4B, a model not listed on the SEA-HELM leaderboard, run with the inference settings of a deployed advisory system on consumer laptop GPUs.

- A reproducible run kit (SEA-HELM commit, recorded local changes, fixed package versions and server settings), with which a second machine reproduced the Sentiment result within one Macro F1 point.

Recommendations

- Score translation with MetricX-24 and summarization with BERTScore when a larger GPU is available, so that NLG results can be compared with the leaderboard.
- Repeat the benchmark with thinking enabled and with full-precision weights, to measure what quantization and the latency-oriented settings cost in Filipino competence.
- Run the baseline Bintu-1 model through the same benchmark, as planned in Chapter IV.
- Add a Taglish (code-switched) evaluation set, since BATAYAN's items are in Filipino only.